

DW_fp_flt2i

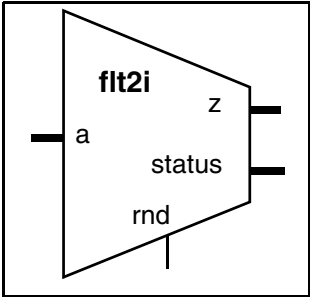
Floating-Point to Integer Converter

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Features and Benefits

- The precision format is parameterizable for either IEEE single, double precision, or a user-defined custom format
- Accuracy conforms to IEEE 754 standard
- DesignWare datapath generator is employed for better timing and area

Revision History



Description

DW_fp_flt2i is a floating-point to integer converter that takes a floating-point number, *a*, to produce an integer number, *z*. The output *z* is always a signed two's complement integer. Component pins are described in [Table 1-1](#) and configuration parameters are described in [Table 1-2](#).

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	(<i>sig_width</i> + <i>exp_width</i> + 1) bits	Input	Floating-point number
rnd	3 bits	Input	Rounding mode; supports all rounding modes described in the Datapath Floating-Point Overview
z	<i>isize</i> bits	Output	Two's complement integer number
status	8 bits	Output	Status flags for the result <i>z</i> For details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> .

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 Default: 23	Word length of fraction field of floating-point number <i>a</i>
exp_width	3 to 31 Default: 8	Word length of biased exponent of floating-point number <i>a</i>
isize	3 to 512 Default: 32	Word length of converted integer number <i>z</i>

Table 1-2 Parameter Description (Continued)

Parameter	Values	Description
ieee_compliance	0 or 1 Default: 0	Level of support for IEEE 754: 0: No support for IEEE 754 NaNs and denormals; NaNs are considered infinities and denormals are considered zeros 1: Fully compliant with the IEEE 754 standard, including support for NaNs and denormals For more, see IEEE 754 Compatibility in the <i>Datapath Floating-Point Overview</i>.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Fast Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_FLT2I_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_flt2i_sim.vhd	VHDL Simulation Model Source Code
dw/dw02/sim_ver/DW_fp_flt2i.v	Verilog Simulation Model Source Code

Table 1-5 Function Example 1 (*sig_width* = 10, *exp_width* = 5, *isize* = 16, *ieee_compliance* = 0)

Description	a (FP format)	rnd	status	z (Integer Number)
Zero	0000_0000_0000_0000	any	0000_0001	0000_0000_0000_0000
Denormal	0000_0000_0000_0001	any	0000_0001	0000_0000_0000_0000
Infinity	0111_1100_0000_0000	any	0110_0000	0111_1111_1111_1111
	1111_1100_0000_0000	any	0110_0000	1000_0000_0000_0000
NaN	0111_1100_0000_0001	any	0110_0000	0111_1111_1111_1111
	1111_1100_0000_0001	any	0110_0000	1000_0000_0000_0000

Table 1-5 Function Example 1 (*sig_width* = 10, *exp_width* = 5, *isize* = 16, *ieee_compliance* = 0) (Continued)

Description	a (FP format)	rnd	status	z (Integer Number)
Normal Number	0110_0011_1111_1111	0	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	1	0010_0000	0000_0011_1111_1111
	0110_0011_1111_1111	2	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	3	0010_0000	0000_0011_1111_1111
	0110_0011_1111_1111	4	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	5	0010_0000	0000_0100_0000_0000
	1110_0011_1111_1111	0	0010_0000	1111_1100_0000_0000

Table 1-6 Function Example 2 (*sig_width* = 10, *exp_width* = 5, *isize* = 16, *ieee_compliance* = 1)^a

Description	a (FP Format)	rnd	status	z (Integer Number)
Zero	0000_0000_0000_0000	any	0000_0001	0000_0000_0000_0000
Denormal	0000_0000_0000_0001	0, 1, 3, 4	0010_1001	0000_0000_0000_0000
	0000_0000_0000_0001	2, 5	0010_0000	0000_0000_0000_0001
	1000_0000_0000_0001	0, 1, 2, 4	0010_1001	0000_0000_0000_0000
	1000_0000_0000_0001	3, 5	0010_0000	1111_1111_1111_1111
Infinity	0111_1100_0000_0000	any	0000_0100	0111_1111_1111_1111
	1111_1100_0000_0000	any	0000_0100	1000_0000_0000_0000
NaN	0111_1100_0000_0001	any	0000_0100	0111_1111_1111_1111
	1111_1100_0000_0001	any	0000_0100	1000_0000_0000_0000
Normal Number	0110_0011_1111_1111	0	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	1	0010_0000	0000_0011_1111_1111
	0110_0011_1111_1111	2	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	3	0010_0000	0110_0011_1111_1111
	0110_0011_1111_1111	4	0010_0000	0000_0100_0000_0000
	0110_0011_1111_1111	5	0010_0000	0110_0011_1111_1111
	1110_0011_1111_1111	0	0010_0000	1111_1100_0000_0000

a. Although the output is an integer number, the tiny bit is set when denormal input is rounded to 0.

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)

The warning messages for this model include the following:

- If an invalid rounding mode has been detected on `rnd`, the following message is displayed:

```
WARNING: <instance_path>:  
        at time = <timestamp>: Illegal rounding mode.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp.all;
-- If using numeric types from std_logic_arith package,
-- comment the preceding line and uncomment the following line:
-- use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_flt2i_inst is
  generic (
    inst_sig_width : POSITIVE := 23;
    inst_exp_width : POSITIVE := 8;
    inst_isize      : INTEGER := 32
  );
  port (
    inst_a      : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_rnd     : in std_logic_vector(2 downto 0);
    z_inst      : out std_logic_vector(inst_isize-1 downto 0);
    status_inst : out std_logic_vector(7 downto 0)
  );
end DW_fp_flt2i_inst;

architecture inst of DW_fp_flt2i_inst is

begin

  -- Instance of DW_fp_flt2i
  U1 : DW_fp_flt2i
  generic map (
    sig_width => inst_sig_width,
    exp_width => inst_exp_width,
    isize => inst_isize
  )
  port map (
    a => inst_a,
    rnd => inst_rnd,
    z => z_inst,
    status => status_inst
  );

end inst;

-- pragma translate_off
configuration DW_fp_flt2i_inst_cfg_inst of DW_fp_flt2i_inst is
  for inst
  end for;

```

```
end DW_fp_flt2i_inst_cfg_inst;  
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_flt2i_inst( inst_a, inst_rnd, z_inst, status_inst );

parameter inst_sig_width = 23;
parameter inst_exp_width = 8;
parameter inst_isize = 32;
parameter inst_ieee_compliance = 0;

input [inst_sig_width+inst_exp_width : 0] inst_a;
input [2 : 0] inst_rnd;
output [inst_isize-1 : 0] z_inst;
output [7 : 0] status_inst;

    // Instance of DW_fp_flt2i
    DW_fp_flt2i #(inst_sig_width, inst_exp_width, inst_isize, inst_ieee_compliance) U1
    (
        .a(inst_a),
        .rnd(inst_rnd),
        .z(z_inst),
        .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
July 2020	DWBB_201912.5	- Adjusted the description of the <i>ieee_compliance</i> parameter in Table 1-2 on page 1 - Added “Suppressing Warning Messages During Verilog Simulation” on page 4
December 2019	DWBB_201912.0	- For STAR 9001572518, corrected status bits for infinity and NAN entries in Table 1-6 on page 3 - Added this Revision History table and the document links on this page

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